Joint Mean-Variance Inference for Nonlinear Mixed-Effects Models with Measurement Errors and Outliers

March, 2022

Ye et al.

1 Introduction

Nonlinear mixed-effects (NLME) models are widely used in many biological applications, such as pharmacokinetic analysis, HIV viral dynamics, and studies of growth and decay. Many NLME models are derived based on the underlying data-generation mechanisms, so they are also called *scientific models* or *mechanistic models*. They may be more desirable than empirical models such as linear mixed effects (LME) models and nonparametric mixed models, since they may provide more reliable predicted values for (unobserved) missing/censored values or future values. In many longitudinal studies, the within-individual repeated measurements exhibit large variations and these variations may change over time. It is important to understand the nature of the within-individual systematic and random variations in order to provide good predictions and efficient statistical inference.

Figure 1 shows the profiles of HIV viral load (HIV-1 RNA copies) longitudinal measurements on a sub-sample of individuals from an AIDS study during an anti-HIV treatment. We see that viral loads decline rapidly in the initial period (first few weeks), roughly following an exponential pattern. The variations of the within-individual repeated measurements appear to *change* over time. Moreover, there seem some *outliers* in the data, i.e., some data do not seem to follow the general patterns of the whole data. It is interesting to see if these large and time-varying within-individual variations may be partially explained by covariates such as CD4 cell counts. Based on some biological arguments, as well as a set of differential equations which describe the viral production and elimination processes, Wu and Ding (1999) showed that the viral load trajectories during the treatment may well be modelled by an NLME model.

/home/lang/Figures/ch9outlier.pdf

Figure 1.1: *Viral loads trajectories (in log_{10} scale) during an anti-HIV treatment.*

To see if the large and time-varying within-individual variations in longitudinal viral load data may be partially explained by time-varying covariates such as CD4 cell counts, we may model the within-individual variations with time-varying covariates and incorporate the large within-individual variations through random effects in the variance model. In other words, we may jointly model the mean and variance of the longitudinal viral load data: we model the mean based on an NLME model and model the variance using another mixed effects model. The advantages of modelling the within-individual variations, compared to NLME models without modelling the variance, are as follows: (a) we may gain good understanding of the nature of the within-individual variation and if this variation may be partially explained by covariates; (b) we may obtain more efficient estimates of the main parameters of interest; (c) as will be described in Section 2, modeling variance also allows us to conduct robust inference to address outliers in the data; and (d) we may provide more accurate predictions for the missing/censored response values or future values.

For example, by modelling the variance, we may obtain an more efficient estimate of initial viral decay rates, which allows us to better evaluate and compare different treatments since the initial viral decay rates may reflect the efficacies of the treatments (Wu and Ding, 1999). Understanding the nature of the within-individual variations in viral load also allows us to choose good timing for treatment interruption.

There is some literature on joint modelling mean and variance for longitudinal data (e.g., Lin et al. 1997; Pourahmadi 1999; Hedeker et al. 2008; German et al. 2022). To our knowledge, all previous work focused on *linear* models, either with random effects or without random effects. In this article, we extend previous work to NLME models, and we also address covariate measurement errors and outlying responses. NLME models are *nonlinear* in the random effects and the dimension of the random effects may be high, so computation for the joint model can be challenging. In this article, we propose a computationally efficient approximate method for approximate likelihood inference.

The article is organized as follows ...

2 The Models

2.1 The mean and variance models

Let y_{ij} be the response value for individual i at time t_{ij}, and let $\mathbf{y}_i = (y_{i1}, y_{i2}, \cdots, y_{in_i})^T$, $i = 1, 2, \cdots, n;\ j = 1, 2, \cdots, n_i$. We consider the following nonlinear mixed effects (NLME) model

$$\mathbf{y}_i = g(\mathbf{x}_i, \boldsymbol{\beta}, \mathbf{u}_i) + \mathbf{e}_i, \qquad i = 1, 2, \cdots, n;\ j = 1, 2, \cdots, n_i, \tag{2.1}$$

where $g(\cdot)$ is a known nonlinear function for the mean $E(\mathbf{y}_i)$, $\mathbf{x}_i$ contains covariates including time, vector $\boldsymbol{\beta} = (\beta_1, \beta_2, \cdots, \beta_p)$ contains (fixed) mean parameters, vector $\mathbf{u}_i$ contains random effects for the mean function, and $\mathbf{e}_i = (e_{i1}, e_{i2}, \cdots, e_{in_i})^T$ are within-individual errors. We assume that $\mathbf{u}_i \sim N(0, D)$, and $e_{i1}, \cdots e_{in_i}$ are conditionally independent given $\mathbf{u}_i$, with $e_{ij} \sim N(0, \sigma_{ij}^2)$. Here we allow the variances of the within-individual random errors, σ_{ij}^2, to be individual-specific and time-dependent. This allows great flexibility in modeling complicated longitudinal data, as discussed in the previous section. Let $\boldsymbol{\sigma}_i = (\sigma_{i1}^2, \cdots, \sigma_{in_i}^2)$.

To understand the nature of the within-individual variations and check if there are some systematic components in these variations, we investigate if the within-individual variances σ_{ij}^2 may be partially explained by time-varying covariates (say) $\mathbf{z}_i = (z_{i1}, \cdots, z_{in_i})$, where z_{ij} is the value of covariate z for individual i at time t_{ij}. Note that covariate $\mathbf{z}_i$ may be different from covariates $\mathbf{x}_i$. For simplicity, here we only consider one covariate, but the method can be easily extended to more than one covariates. Moreover, the within-individual variations σ_{ij}^2 may vary substantially across individuals. Thus, we model the within-individual variations and introduce additional random effects to account for the associations within an individual and variations across individuals, i.e., we consider the following model for the variance

$$\log(\sigma_{ij}^2) = \alpha_1 + \alpha_2 z_{ij} + \eta a_i, \qquad i = 1, \cdots, n, \quad j = 1, 2, \cdots, n_i, \tag{2.2}$$

where vector $\boldsymbol{\alpha} = (\alpha_1, \alpha_2)$ and η are fixed parameters, and a_i is a random effect for the variance model. We may assume that $\exp(a_i)$ follows an inverse gamma distribution or $a_i \sim N(0, 1)$. In the variance model (2.2), we model the log-transformation of the variances, $\log(\sigma_{ij}^2)$, since the variances σ_{ij}^2 are strictly positive and the range of the log-transformation $\log(\sigma_{ij}^2)$ is $(-\infty, \infty)$, which allows us to introduce arbitrary covariate in the model.

In the above mean and variance models, we jointly model the conditional mean $E(\mathbf{y}_i|\mathbf{u}_i) = g(\mathbf{x}_i, \boldsymbol{\beta}, \mathbf{u}_i)$ and the conditional variance $Var(\mathbf{y}_i|\mathbf{u}_i) = diag(\sigma_{i1}^2, \cdots, \sigma_{in_i}^2)$. Modelling the variances allows for more efficient estimation of the mean parameters and better understanding of the within-individual variabilities.

2.2 Robust model

In the variance model (2.2), if we assume that $\exp(a_i) \sim k/\chi_k^2$, i.e., the inverse χ^2 distribution, which is a special case of a inverse gamma distribution, then the random errors e_{ij} for the within-individual repeated measurement follow a $t(k)$-distribution with degrees of freedom k (Lang et al. 1989). The $t(k)$-distribution has heavier tails than the normal distribution, and thus it incorporates possible outliers in the within-individual repeated measurements. The smaller the degrees of freedom k, the heavier the tails. The $t(k)$-distribution reduces to the standard normal distribution when $k \to \infty$. Therefore, in this case models (2.1) and (2.2) may be used for robust inference.

Similarly, we may allow the variances for the random effects for the mean model $\mathbf{u}_i$ to be individual-specific, i.e., $\mathbf{u}_i \sim N(0, D_i)$, and model the covariance matrix D_i in a similar way as σ_{ij}^2. While the approaches are conceptually similar, for simplicity we focus on the variance model (2.2) and assume $D_i = D$ in this paper.

2.3 Covariate measurement errors

The covariate z_{ij} in the variance model may be measured with errors, such as CD4 cell count. The covariate effect on the variance, if any, may be masked if the measurement errors are not addressed. To address the measurement errors, we may view the repeated measurements of the covariate as its "replicates" in order to estimate the magnitude of the measurement errors and model the covariate process using an empirical

LME model.

Let z_{ij}^* be the unobserved true covariate value for individual i at time t_{ij} and let z_{ij} be the corresponding observed value with measurement error. We may consider the following mixed effects model as a classic measurement error model

$$z_{ij} \equiv z_{ij}^* + \epsilon_{ij} = q(t_{ij}, \boldsymbol{\gamma}, \mathbf{b}_i) + \epsilon_{ij}, \tag{2.3}$$

where $q(\cdot)$ is a known linear function or a unknown smooth function, $\boldsymbol{\gamma}$ contains fixed parameters, $\mathbf{b}_i$ contains random effects, and ϵ_{ij} is measurement error. We assume that $b_i \sim N(0, B)$, and independently ϵ_{ij} $i.i.d.$ $\sim N(0, \omega^2)$.For example, we may model the CD4 process using a quadratic mixed effects model $z_{ij} = \gamma_0 + \gamma_1 t_{ij} + \gamma_2 t_{ij}^2 + b_i + \epsilon_{ij}$. Alternatively, for more complex CD4 trajectories, we may choose $q(t_{ij}, \boldsymbol{\gamma}, \mathbf{b}_i)$ to be a nonparametric smooth function, then approximate it using a basis-based approach, which leads to an approximate LME model (Wu, 2009). In either case, we may consider an LME model for the measurement error model (2.3).

3 Approximate Likelihood Inference

Let $\boldsymbol{\theta} = (\boldsymbol{\beta}, \boldsymbol{\gamma}, \boldsymbol{\xi})$ denote the collection of all fixed parameters in the models, where $(\boldsymbol{\beta}, \boldsymbol{\gamma})$ contains fixed mean parameters, and $\boldsymbol{\xi} = (\boldsymbol{\alpha}, D, \eta, \omega^2)$ contains fixed variance (dispersion) parameters. Let $\mathbf{v}_i = (\mathbf{u}_i, a_i, \mathbf{b}_i)$ denote the collection of all random effects, and let $\mathbf{v} = (\mathbf{v}_1, \cdots, \mathbf{v}_n)$. Standard likelihood inference for $\boldsymbol{\theta}$ is based on the following (marginal) likelihood function, which integrates out the unobservable random effects

$$L(\boldsymbol{\theta}) = \prod_{i=1}^{n} \int \int \int f(\mathbf{y}_i |\, \mathbf{x}_i, \mathbf{z}_i, \boldsymbol{\beta}, D, \sigma_i^2, \mathbf{u}_i) f(\sigma_i^2 | \boldsymbol{\alpha}, a_i) f(\mathbf{z}_i | \boldsymbol{\gamma}, \mathbf{b}_i) f(\mathbf{u}_i, \mathbf{b}_i, a_i | \boldsymbol{\theta})\, d\mathbf{u}_i\, da_i\, d\mathbf{b}_i,$$

where $f(\cdot)$ denotes a generic density function. The above likelihood involves a high-dimensional and intractable integration. Monte Carlo EM algorithms or Gaussian quadrature numerical integration methods can be computationally very intensive. Thus, in the following, we discuss a computationally more efficient approximate method for approximate likelihood inference, using the h-likelihood (Lee et al. 2017) or conditional second-order generalized estimating equations (CGEE2, Vonesh et al. 2002).

The h-loglikelihood (Lee et al. 2017) can be written as

$$h(\boldsymbol{\theta}, \mathbf{v}) = \sum_{i=1}^{n} \left\{ \log f(\mathbf{y}_i | \, \mathbf{x}_i, \mathbf{z}_i, \boldsymbol{\beta}, D, \sigma_i^2, \mathbf{u}_i) + \log f(\sigma_i^2 | \boldsymbol{\alpha}, a_i) + \log f(\mathbf{z}_i | \boldsymbol{\gamma}, \mathbf{b}_i) + \log f(\mathbf{u}_i, \mathbf{b}_i, a_i | \boldsymbol{\theta}) \right\}$$

where the random effects $\mathbf{v}_i = (\mathbf{u}_i, \mathbf{b}_i, a_i)$ are treated as "parameters". Given parameter $\boldsymbol{\theta}$, estimates of the random effects $\mathbf{v}$ can be obtained by maximizing the h-loglikelihood $h(\boldsymbol{\theta}, \mathbf{v})$ with respect to $\mathbf{v}$, denoted by $\hat{\mathbf{v}}$, i.e., $\hat{\mathbf{v}}$ solves $\partial h(\boldsymbol{\theta}, \mathbf{v})/\partial \mathbf{v} = 0$. Then, we can obtain estimates of the parameters $\boldsymbol{\theta}$ by maximizing $h(\boldsymbol{\theta}, \hat{\mathbf{v}})$. The process iterates until convergence. For binary or discrete response $\mathbf{y}_i$, Lee et al. (2017) recommends adding an adjusted term to reduce bias and obtain estimates of the parameters $\boldsymbol{\theta}$ by maximizing the following "adjusted profile likelihood"

$$\log L(\boldsymbol{\theta}) \approx p_{\mathbf{v}}(h) \equiv \left[h(\boldsymbol{\theta}, \mathbf{v}) - \frac{1}{2} \log det\{H(h; \mathbf{v})/(2\pi)\} \right] \Bigg|_{\mathbf{v}=\hat{\mathbf{v}}},$$

where the second term is an adjusted term, and $H(h; \mathbf{v}) = -\partial^2 h/\partial \mathbf{v}^2$ and $\hat{\mathbf{v}}$ solves $\partial h/\partial \mathbf{v} = 0$. Lee and Nelder (2001) show that the h-likelihood estimate is the first-order Laplace approximation to the marginal log-likelihood $\log L(\boldsymbol{\theta})$.

Vonesh et al. (2002) proposed the CGEE2 method which also maximizes $h(\boldsymbol{\theta}, \mathbf{v})$ but without the adjusted term. That is, the CGEE2 estimate may be viewed as the h-likelihood estimate without the adjusted term. They showed that their CGEE2 estimate is asymptotically equivalent to the h-likelihood estimate, the linearization estimate of Lindstrom and Bates (1990) for NLME models, and the penalized quasi-likelihood (PQL) estimate of Breslow and Clayton (1993), when both the sample size n and the numbers of repeated measurements $\min_i\{n_i\}$ go to ∞. Moreover, they showed that the CGEE2 estimates are consistent and asymptotically normally distributed. Following their ideas, we propose the following iterative algorithm to obtain approximate MLE of the parameter $\boldsymbol{\theta} = (\boldsymbol{\beta}, \boldsymbol{\gamma}, \boldsymbol{\xi})$. Starting with some initial values $(\boldsymbol{\theta}^{(0)}, \mathbf{v}^{(0)})$, at iterative $k, k = 0, 1, 2, \cdots,$

- Step 1: Solve equation $\partial h(\boldsymbol{\theta}^{(k)}, \mathbf{v})/\partial \mathbf{v} = 0$ and obtain updated estimate of the random effects $\hat{\mathbf{v}}^{(k+1)}$;

- Step 2: Solve equation $\partial h(\boldsymbol{\beta}, \boldsymbol{\gamma}, \boldsymbol{\xi}^{(k)}; \mathbf{v}^{(k+1)})/\partial(\boldsymbol{\beta}, \boldsymbol{\gamma}) = 0$ and obtain updated esti-

mates of the mean parameters $(\boldsymbol{\beta}^{(k+1)}, \boldsymbol{\gamma}^{(k+1)})$;

- Step 3: Solve equation $\partial h(\boldsymbol{\beta}^{(k+1)}, \boldsymbol{\gamma}^{(k+1)}, \boldsymbol{\xi}; \mathbf{v}^{(k+1)})/\partial \boldsymbol{\xi} = 0$ and obtain updated estimates of the variance-covariance parameters $\boldsymbol{\xi}^{(k+1)}$.

Iterate the above steps until convergence, we obtain approximate MLE $\hat{\boldsymbol{\theta}}$. Standard errors of the estimates may be obtained from the diagonal elements (square roots) of the inverse of the Fisher information matrix.

4 Real Data Examples

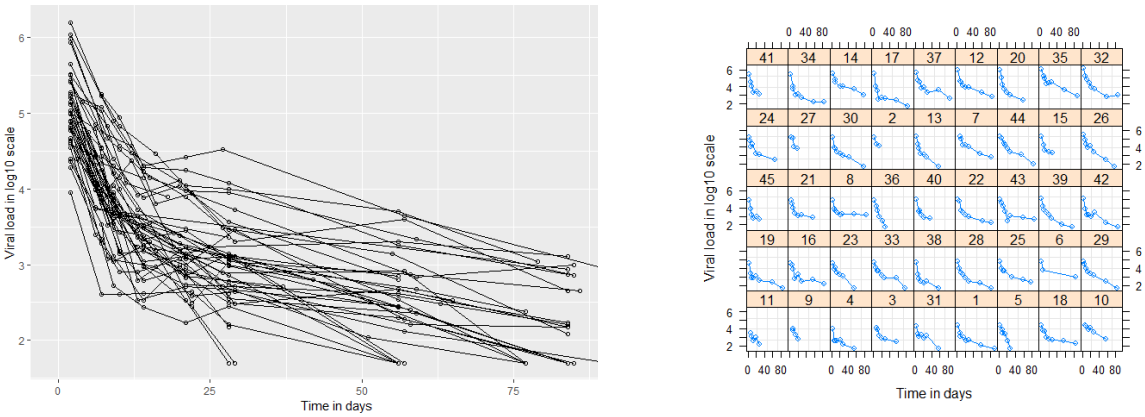
Analyze a real HIV/AIDS dataset using three methods: the Linstrom and Bates (1990) method implemented in R package *nlme* without modeling the variances, the two-step method, and the proposed method. The results should show efficiency gain (smaller standard errors) of the estimates $\hat{\boldsymbol{\beta}}$ which allow us to compare different treatments more effectively. Estimates of $\boldsymbol{\alpha}$ also allow us to see if CD4 partially explains the within-individual variance σ_{ij}^2.

4.1 S315 Data Set

In this section, we analyze the S315 data set from the AIDS study mentioned in Section 1, based on the proposed models and inference methods. The data set recorded the longitudinal measurements of HIV viral load, as well as other biomarkers (e.g. CD4 cell counts), on 45 HIV-infected subjects during the first 3 months of an anti-HIV treatment.... (some background about the study). In total, 311 repeated measurements of viral load were collected over time, with the median number being 7 per subject (ranging from 3 to 8). Moreover, the associated CD4 measures had about 20% data missing (63 out of 311) and was prone to measurement errors. The objective of the data analysis is to study the viral load dynamic over time with considering: 1) the large and time-varying within-individual variation, 2) measurement errors and missing data in the CD4 measures, and 3) potential outliers in the viral load data.

Figure 4.1 shows the longitudinal trajectories of viral load of all subjects (the trajectories based on a sub-samples was shown in Figure 1.1). As discussed in Section 1, we see that viral loads decline rapidly in the initial stage of the treatment, so we con-

Figure 4.1: Viral load trajectories (in $\log_{10}$ scale) during the first 3 months of an anti-HIV treatment: all subjects



(a) All trajectories in one panel

(b) Trajectory for each subject

sidered a one-compartment exponential viral decay model proposed by Wu and Ding (1999) for the the viral dynamics. The variations of the within-individual repeated viral loads appear to be large and not constant over time and across individuals. To check if there are some systematic components in these variations, we modeled the within-individual variances in terms of the time-varying CD4 measures and a random effect. The measurement errors and missing data in the CD4 measures were addressed by assuming a LME measurement error model, which can be used to estimate the magnitude of the measurement error, as well as "fill in" the missing data in the modelled covariate process. Moreover, based on the discussion in Section 2, the proposed models are robust to the pontential outliers in viral loads if assuming an inverse χ^2 distribution to the random effect in the model for the within-individual variances.

Models

To model the viral load profile during the early stage (the first 3 months) of an anti-HIV treatment, we considered the following NLME model (Wu and Ding (1999))

$$y_{ij} = (\beta_1 + u_{i1}) + \beta_2 e^{-(\beta_3 + u_{i3})t_{ij}} + e_{ij}, \tag{4.1}$$

where y_{ij} is the $\log_{10}$-transformed viral load measured at time t_{ij} for individual i, β_1 and β_2 are baseline viral load values, β_3 is the decay rate, and e_{ij} is the within-individual random error with its variance assumed to be time-dependent and individual-specific,

that is, $e_{ij} \sim \mathcal{N}(0, \sigma_{ij}^2)$. The model only contains the random effects u_{i1} and u_{i3} for the baseline parameter β_1 and the decay rate β_3, respectively, as this is the optimal setting considering both the model complexity and goodness of fit (AIC value). We assume the random effects $u_i = (u_{i1}, u_{i3})$ follows a multivariate normal distribution with zero means and a fixed variance matrix D. The $\boldsymbol{\beta} = (\beta_1, \beta_2, \beta_3)$ are fixed parameters representing the trajectory at the population level and the random effects u_i characterize the individual-specific effects and the between-individual variations.

In addition to the mean model in (4.1), we considered a variance model to examine if the within-individual variances could be partially explained by the CD4 measures. Given z_{ij}^* denotes the "true" but unobservable (log-) CD4 cell count at time t_{ij} for individual i, the variance model is written as

$$\log(\sigma_{ij}^2) = \alpha_1 + \alpha_2 z_{ij}^* + a_i, \tag{4.2}$$

where $\boldsymbol{\alpha} = (\alpha_1, \alpha_2)$ are fixed variance parameters and a_i is a random effect following a distribution of $\mathcal{N}(0, \sigma_a^2)$ or an inverse-χ^2 distribution if addressing outlier is needed. We additionaly expect more efficient estimation of the mean parameters in the mean model (4.1) by modelling the variances.

It is worth mentioning that model (4.2) links the within-individual variance to the unobserved true value of covariate z_{ij}^* rather than the observed value z_{ij}. In practice, the longitudinal clinic data are often measured with measurement errors. Directly using the observed data with ignoring the measurement errors may lead to misleading statistical inference. We can model the CD4 covariate process to partially address the measurement errors. Based on the CD4 growth trajectories (see Figure A.1 and A.2) and AIC criteria, the following LME model was selected for the data set

$$z_{ij} = z_{ij}^* + \varepsilon_{ij} \equiv \gamma_0 + \gamma_1 t_{ij} + \gamma_2 t_{ij}^2 + b_i + \epsilon_{ij}, \tag{4.3}$$

where the unobserved true CD4 covariate process is assumed to be $z_{ij}^* = \gamma_0 + \gamma_1 t_{ij} + \gamma_2 t_{ij}^2 + b_i$, b_i is a random effect, and ϵ_{ij} stands for the measurement error. We assume $b_i \sim \mathcal{N}(0, \sigma_b^2)$, $\epsilon_{ij} \sim_{iid} \mathcal{N}(0, \sigma_z^2)$ and they are independent from each other.

Table 4.1: Analyses of the AIDS data under different models.

Model		Mean parameters								Variance parameters				
		β_1	β_2	β_3	α_1	α_2	γ_0	γ_1	γ_2	σ_a	σ_b	σ_z	d_1	d_3
Standard	Est	2.32	2.99	5.95									0.32	2.16
	SE	0.070	0.065	0.453										
	p-val	<.001	<.001	<.001										
TS	Est	2.52	2.97	7.53	-20.94	3.34	5.20	1.64	-1.28	1.08	0.37	0.23	0.34	1.39
	SE	0.062	0.045	0.338	1.210	0.219	0.061	0.207	0.228					
	p-val	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001					
JM	Est	2.52	2.96	7.45	-15.46	2.33	5.20	1.58	-1.15	0.83	0.37	0.20	0.33	1.44
	SE.bt	0.066	0.044	0.412	1.953	0.356	0.055	0.162	0.183					
	p-val	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001					
Robust JM	Est	2.53	2.97	7.60	-18.87	2.96	5.20	1.55	-1.12		0.37	0.20	0.33	1.43
	SE.bt	0.066	0.043	0.400	2.171	0.396	0.059	0.171	0.178					
	p-val	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001					

Standard: the linearization method based on Linstram and Bates (1990). TS: two-step method. JM: Joint model. Robust JM: Joint model with an inverse-χ^2 (k=3) distribution for a_i
d_1 and d_3 are the standard deviations of random effects u_{i1} and u_{i3}, respectively

Inference Methods

The proposed h-likelihood inference method was used to jointly estimate parameters in model (4.1), (4.2) and (4.3). We also considered two more approaches to analyze the data for comparison: i) a standard approach (Lindstrom and Bates (1990)) without modeling the variances that is implemented in the R package `nlme`; ii) a two-step approach which fits the LME covariate model (4.3) separately and estimates the unobserved true CD4 values in the first step and substitutes the unobserved true CD4 covariate in the variance model (4.2) by their estimates in the second step. The LME model fitting and estimation in the first step was implemented by using the package `nlme`, while in the second step, the mean and variance models were jointly estimated using the proposed h-likelihood inference method. Compared to the joint inference for the three models simultaneously, the two-step method may underestimate the standard errors of the parameter estimates in the variance model (4.2) as it treats the estimated values as if they were observed and thus does not incorporate the estimation uncertainty. The standard approach would be less efficient as the variances were not modelled.

Parameter Estimates

Table 4.1 summarizes estimation results from the four above-mentioned approaches.

Describe, interpret, and discuss, ...

Model diagnostic ...

4.2 DTP Data Set

Describe data set: 1) sample size 2) different ARV initiation treatment 3) outliers

Table 4.2: Analyses of the DTP data under different models.

Model		Mean parameters								Variance parameters						
		β_1	β_2	β_3	α_1	α_2	γ_0	γ_1	γ_2	σ_a	σ_{b0}	σ_{b1}	σ_{b2}	σ_z	d_2	d_3
Standard	Est	1.50	2.00	6.61											1.40	5.77
	SE	0.006	0.038	0.186												
	p-val	<.001	<.001	<.001												
TS	Est	1.60	4.03	35.23	3.14	-1.05	5.92	0.63	-0.34	1.50	0.76	1.23	0.99	0.26	0.71	26.86
	SE	0.001	0.018	0.634	0.137	0.022	0.019	0.047	0.044							
	p-val	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001							
Robust JM	Est	1.53	2.23	9.63	1.46	-0.89	5.90	0.73	-0.44		0.61	0.57	0.86	0.24	1.33	6.11
	SE	0.001	0.032	0.145	0.128	0.021	0.016	0.032	0.036							
	p-val	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001							

Standard: the linearization method based on Linstram and Bates (1990). TS: two-step method. JM: Joint model. Robust JM: Joint model with an inverse-χ^2 (k=3) distribution for a_i

d_2 and d_3 are the standard deviations of random effects u_{i2} and u_{i3}, respectively

Table 4.3: Compare the viral load dynamics between two regimen groups.

Model		IIN/ENH							PI/IIN/ENH/							
		β_1	β_2	β_3	α_1	α_2	γ_0	γ_1	β_1	β_2	β_3	α_1	α_2	γ_0	γ_1	γ_2
Standard	Est	1.61	2.70	23.38					1.57	3.77	25.95					
	SE	0.027	0.154	2.166					0.010	0.110	1.247					
	p-val	<.001	<.001	<.001					<.001	<.001	<.001					
TS	Est	1.54	2.71	30.05	11.31	-2.41	6.08	0.22	1.54	3.88	29.78	-2.90	-0.28	6.20	1.08	-0.83
	SE	0.005	0.056	1.451	0.699	0.113	0.062	0.043	0.003	0.091	1.400	0.623	0.097	0.045	0.111	0.119
	p-val	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	0.003	<.001	<.001	<.001
Robust JM	Est	1.54	2.67	31.33	14.01	-2.89	6.06	0.28	1.54	3.88	29.73	-2.95	-0.28	6.20	1.11	-0.89
	SE	0.003	0.047	1.590	0.744	0.121	0.061	0.032	0.003	0.091	1.396	0.582	0.090	0.046	0.083	0.087
	p-val	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	0.002	<.001	<.001	<.0001

Standard: the linearization method based on Linstram and Bates (1990). TS: two-step method. JM: Joint model. Robust JM: Joint model with an inverse-χ^2 (k=3) distribution for a_i

Table 4.4: Compare the viral load dynamics between two regimen groups.

Model		IIN														
		β_1	β_2	β_3	α_1	α_2	γ_0	γ_1	β_1	β_2	β_3	α_1	α_2	γ_0	γ_1	γ_2
Standard	Est	1.59	3.14	25.80												
	SE	0.015	0.076	1.142												
	p-val	<.001	<.001	<.001												
TS	Est	1.55	2.49	25.62	3.28	-1.18	6.00	0.31								
	SE	0.003	0.028	0.747	0.293	0.048	0.045	0.032								
	p-val	<.001	<.001	<.001	<.001	<.001	<.001	<.001								
Robust JM	Est	1.54	2.42	25.04	3.45	-1.21	5.99	0.34								
	SE	0.003	0.027	0.732	0.290	0.047	0.046	0.026								
	p-val	<.001	<.001	<.001	<.001	<.001	<.001	<.001								

Standard: the linearization method based on Linstram and Bates (1990). TS: two-step method. JM: Joint model. Robust JM: Joint model with an inverse-χ^2 (k=3) distribution for a_i

5 Simulation Study

In the simulation study, we will evaluate the proposed method and compare it with the Linstrom and Bates (1990) method implemented in R package *nlme* without modeling the variances and the two-step method. The simulation will be designed in the following ways: i) compare estimates, standard errors, MSEs, and coverage probabilities; ii) bootstrap standard errors (either parametric or nonparametric bootstrap);

iii) robustness of the proposed method (with inverse-gamma distribution for a_i) for simulated data with 5% outliers (adding a large number, say 10, to 5% simulated data); iv) different sample size n (say $n = 100, 200$) and different values of n_i (say, 15, 30).

i & ii. Compare estimates, standard errors, MSEs, and coverage probabilities, with parametric bootstrap standard errors

Data are simulated for $n = 100$ subjects, with $n_i = 15$ repeated measurements for each subject. The entire study period is set to be $[0, 1]$, and the 15 measuring time points are equally allocated within the time interval. The true parameter values are chosen based on the estimates from the empirical analysis in Section 4. Specifically,

$$\begin{aligned}
\boldsymbol{\beta} &= (\beta_1, \beta_2, \beta_3)^T = (2.5, 3.0, 7.5)^T, \\
\boldsymbol{\alpha} &= (\alpha_0, \alpha_1)^T = (-8, 1.5)^T, \\
\boldsymbol{\gamma} &= (\gamma_0, \gamma_1, \gamma_2)^T = (5.2, 1.6, -1.2)^T, \\
(d_1, d_3) &= (0.4, 1.0), \quad \sigma_a = 0.7, \quad \sigma_b = 0.4, \quad \sigma_z = 0.2.
\end{aligned}$$

The structure of the variance-covariance matrix for the random effects $\boldsymbol{v}_i = (\boldsymbol{u}_i, a_i, b_i)$ varies depending on different assumptions. For now, we assume the random effects are correlated within the same model and independent across different models. The starting values for the inference are set to be the true values. The number of simulation repetition is set to be 100.

Table 5.1 reports "Est", "rBias", "rMSE", "SE.em", "SE", and "Coverage" of the parameters in models. For each parameter, "Est" is the average of the 100 estimates from the 100 simulation runs, "rBias" and "rMSE" are the relative bias and mean square error of the 100 parameter estimates, which are defined as follows:

$$\begin{aligned}
\text{rBias (\%) of } \hat{\beta} &= \left| \left(\frac{1}{M} \sum_{m=1}^{M} \hat{\beta}^{(m)} - \beta \right) / \beta \right| \times 100\%, \\
\text{rMSE (\%) of } \hat{\beta} &= \left| \frac{1}{M} \sum_{m=1}^{M} \left(\hat{\beta}^{(m)} - \beta \right)^2 / \beta \right| \times 100\%,
\end{aligned}$$

where $\hat{\beta}^{(m)}$ is the estimate of β in the m-th simulation run and M is the total number of simulation repetitions, "SE.em" is the empirical standard deviation of the 100 estimates,

"SE" is the square root of the average of the 100 variance estimates, and "Coverage" is the empirical coverage of the true parameter value among the 100 95% confidence intervals from the simulation runs.

In addition, **parametric** bootstrapping method was used to adjust the SE for the parameter estimates. We conducted 100 bootstrapping runs for each of 100 simulation runs. "SE.BT" is the average of bootstrapping SE from the 100 simulation runs and " Coverage.bt" is the corresponding coverage rate. The bootstrapping SE was calculated as the empirical SE of the estimates from the 100 bootstrapping runs.

Table 5.1: Simulation results

Model	Parameter	β_1	β_2	β_3	α_1	α_2	γ_0	γ_1	γ_2
	True value	2.5	3.0	7.5	-8	1.5	5.2	1.6	-1.2
Standard	Estimate	2.527	2.996	8.054					
	SE.mo	0.067	0.148	0.862					
	SE.em	0.072	0.116	0.880					
	rBias(%)	1.076	0.135	7.393					
	rMSE(%)	0.235	0.441	14.313					
	Coverage	0.920	0.990	0.910					
TS	Estimate	2.509	2.994	7.699	-8.392	1.547	5.206	1.598	-1.198
	SE.mo	0.052	0.069	0.433	0.492	0.087	0.042	0.063	0.061
	SE.em	0.050	0.082	0.486	0.725	0.130	0.041	0.058	0.056
	rBias(%)	0.375	0.190	2.655	4.894	3.106	0.120	0.144	0.167
	rMSE(%)	0.102	0.222	3.647	8.426	1.252	0.033	0.207	0.258
	Coverage	0.960	0.920	0.930	0.760	0.800	0.980	0.970	0.960
JM	Estimate	2.511	2.995	7.759	-8.069	1.489	5.205	1.603	-1.202
	SE.mo	0.052	0.069	0.441	0.486	0.086	0.042	0.060	0.058
	SE.em	0.051	0.082	0.487	0.774	0.138	0.041	0.056	0.054
	SE. bt	0.057	0.075	0.495	0.690	0.122	0.043	0.061	0.059
	rBias(%)	0.428	0.174	3.455	0.866	0.729	0.091	0.210	0.149
	rMSE(%)	0.106	0.222	4.027	7.479	1.267	0.033	0.193	0.237
	Coverage	0.950	0.930	0.920	0.800	0.810	0.980	0.970	0.960
	Coverage.bt	0.970	0.960	0.950	0.930	0.920	0.970	0.960	0.970

Standard: the linearization method based on Linstram and Bates (1990). TS: two-step method. JM: Joint model. SE.mo: model-based SE. SE.em: empirical SE from simulation. SE.bt: Bootstrap SE.

iii. Robust modeling setting (assuming inverse χ^2 distribution for a_i)

We consider two values, i.e. $k = 3$ and $k = 5$, for the degree of freedom of the inverse χ^2 distribution.

(a) Outliers naturally from data generation, $k = 3$

(b) Outliers naturally from data generation, $k = 5$

Table 5.2: Inverse χ^2 distribution with 3 degree of freedom for a_i: both data generation and modeling

Model	Parameter	β_1	β_2	β_3	α_1	α_2	γ_0	γ_1	γ_2
	True value	2.5	3.0	7.5	-8	1.5	5.2	1.6	-1.2
Standard	Estimate	2.490	3.007	7.546					
	SE.mo	0.090	0.222	1.245					
	SE.em	0.104	0.171	1.269					
	rBias(%)	0.405	0.248	0.612					
	rMSE(%)	0.431	0.963	21.293					
	Coverage	0.920	0.990	0.950					
TS	Estimate	2.505	2.991	7.582	-8.024	1.552	5.199	1.605	-1.204
	SE.mo	0.056	0.081	0.500	0.491	0.087	0.042	0.063	0.061
	SE.em	0.064	0.085	0.648	0.750	0.134	0.048	0.064	0.063
	rBias(%)	0.191	0.302	1.090	0.304	3.436	0.025	0.310	0.317
	rMSE(%)	0.162	0.242	5.636	6.973	1.368	0.043	0.254	0.328
	Coverage	0.920	0.940	0.850	0.810	0.820	0.940	0.960	0.960
JM	Estimate	2.507	2.991	7.645	-7.738	1.500	5.197	1.611	-1.208
	SE.mo	0.055	0.081	0.507	0.486	0.086	0.042	0.060	0.058
	SE.em	0.064	0.085	0.659	0.806	0.144	0.048	0.062	0.061
	SE.bt	0.071	0.106	0.679	0.755	0.134	0.042	0.061	0.058
	rBias(%)	0.266	0.298	1.938	3.275	0.021	0.056	0.709	0.701
	rMSE(%)	0.163	0.241	6.023	8.892	1.363	0.043	0.245	0.316
	Coverage	0.910	0.940	0.850	0.740	0.780	0.940	0.920	0.910
	Coverage.bt	0.960	0.990	0.970	0.910	0.920	0.940	0.920	0.930

Standard: the linearization method based on Linstram and Bates (1990). TS: two-step method. JM: Joint model. SE.mo: model-based SE. SE.em: empirical SE from simulation. SE.bt: Bootstrap SE.

Table 5.3: Inverse χ^2 distribution with 5 degree of freedom for a_i: both data generation and modeling

Model	Parameter	β_1	β_2	β_3	α_1	α_2	γ_0	γ_1	γ_2
	True value	2.5	3.0	7.5	-8	1.5	5.2	1.6	-1.2
Standard	Estimate	2.494	3.007	7.643					
	SE.mo	0.074	0.169	0.936					
	SE.em	0.075	0.127	0.875					
	rBias (%)	0.225	0.239	1.913					
	rMSE (%)	0.224	0.538	10.382					
	Coverage	0.960	1.000	0.980					
TS	Estimate	2.491	3.003	7.466	-8.482	1.607	5.200	1.589	-1.190
	SE.mo	0.056	0.075	0.456	0.489	0.087	0.042	0.063	0.061
	SE.em	0.059	0.105	0.627	1.804	0.320	0.042	0.061	0.061
	rBias (%)	0.377	0.093	0.455	6.020	7.164	0.008	0.686	0.816
	rMSE (%)	0.143	0.367	5.206	43.168	7.516	0.034	0.240	0.320
	Coverage	0.950	0.820	0.850	0.430	0.440	0.950	0.970	0.960
JM	Estimate	2.493	3.003	7.536	-7.787	1.484	5.199	1.595	-1.194
	SE.mo	0.056	0.076	0.464	0.483	0.086	0.042	0.060	0.058
	SE.em	0.059	0.105	0.639	1.478	0.263	0.043	0.062	0.061
	SE.bt	0.068	0.101	0.639	1.552	0.278	0.043	0.061	0.059
	rBias (%)	0.297	0.107	0.478	2.657	1.098	0.023	0.323	0.498
	rMSE (%)	0.141	0.366	5.403	27.599	4.594	0.035	0.240	0.311
	Coverage	0.940	0.820	0.850	0.460	0.480	0.940	0.980	0.960
	Coverage.bt	0.970	0.940	0.930	0.970	0.970	0.940	0.970	0.940

Standard: the linearization method based on Linstram and Bates (1990). TS: two-step method. JM: Joint model. SE.mo: model-based SE. SE.em: empirical SE from simulation. SE.bt: Bootstrap SE.

Table 5.4: 5% outliers and inverse χ^2 distribution with 3 degree of freedom for a_i: modeling only

Model	Parameter	β_1	β_2	β_3	α_1	α_2	γ_0	γ_1	γ_2
	True value	2.5	3.0	7.5	-8	1.5	5.2	1.6	-1.2
Standard	Estimate	2.514	3.001	7.660					
	SE.mo	0.081	0.192	1.072					
	SE.em	0.081	0.156	1.042					
	rBias (%)	0.564	0.027	2.139					
	rMSE (%)	0.268	0.804	14.678					
	Coverage	0.970	0.990	0.930					
TS	Estimate	2.511	2.998	7.639	-7.618	1.494	5.189	1.601	-1.202
	SE.mo	0.056	0.082	0.506	0.487	0.086	0.042	0.063	0.061
	SE.em	0.065	0.104	0.660	1.161	0.207	0.043	0.067	0.065
	rBias (%)	0.439	0.073	1.854	4.772	0.418	0.214	0.088	0.135
	rMSE (%)	0.170	0.358	6.013	18.509	2.844	0.037	0.279	0.343
	Coverage	0.880	0.880	0.890	0.560	0.610	0.940	0.940	0.960
JM	Estimate	2.512	2.998	7.673	-7.186	1.416	5.188	1.606	-1.205
	SE.mo	0.056	0.083	0.512	0.482	0.085	0.043	0.060	0.058
	SE.em	0.064	0.104	0.652	1.341	0.241	0.043	0.066	0.063
	SE.bt	0.073	0.111	0.719	0.745	0.133	0.043	0.061	0.058
	rBias (%)	0.478	0.059	2.308	10.178	5.574	0.236	0.368	0.398
	rMSE (%)	0.168	0.356	6.014	30.528	4.305	0.038	0.274	0.330
	Coverage	0.870	0.890	0.900	0.480	0.480	0.910	0.940	0.950
	Coverage.bt	1.000	0.940	0.950	0.690	0.760	0.940	0.950	0.950

Standard: the linearization method based on Linstram and Bates (1990). TS: two-step method. JM: Joint model. SE.mo: model-based SE. SE.em: empirical SE from simulation. SE.bt: Bootstrap SE.

(c) 5% manually added outlier, $k = 3$

iv. (a) Different sample size: $n = 200$

iv. (b) Different number of repeated measurements: $n_i = 30$

6 Discussion

References

M. Lindstrom and D. Bates. Nonlinear mixed effects models for repeated measures data. *Biometrics*, 46:673–687, 1990.

H. Wu and A. A. Ding. Population hiv-1 dynamics in vivo: Application models and

Table 5.5: Simulation results with $n = 200$

Model	Parameter	β_1	β_2	β_3	α_1	α_2	γ_0	γ_1	γ_2
	True value	2.5	3.0	7.5	-8	1.5	5.2	1.6	-1.2
Standard	Estimate	2.507	2.976	7.425					
	SE.mo	0.048	0.104	0.561					
	SE.em	0.052	0.079	0.615					
	rBias(%)	0.289	0.785	1.000					
	rMSE(%)	0.110	0.223	5.060					
	Coverage	0.920	0.980	0.920					
TS	Estimate	2.503	2.991	7.526	-8.331	1.536	5.199	1.605	-1.203
	SE.mo	0.036	0.048	0.295	0.347	0.061	0.030	0.045	0.043
	SE.em	0.043	0.056	0.393	0.446	0.079	0.028	0.043	0.041
	rBias(%)	0.108	0.295	0.353	4.133	2.407	0.013	0.309	0.281
	rMSE(%)	0.075	0.108	2.047	3.833	0.500	0.015	0.114	0.140
	Coverage	0.920	0.910	0.860	0.750	0.820	0.950	0.960	0.970
JM	Estimate	2.504	2.992	7.582	-8.003	1.478	5.198	1.611	-1.207
	SE.mo	0.036	0.048	0.299	0.343	0.060	0.030	0.042	0.041
	SE.em	0.044	0.057	0.394	0.544	0.096	0.028	0.042	0.040
	SE.bt	0.040	0.052	0.340	0.470	0.083	0.030	0.043	0.041
	rBias(%)	0.145	0.257	1.094	0.043	1.493	0.043	0.670	0.604
	rMSE(%)	0.076	0.108	2.138	3.667	0.640	0.015	0.115	0.135
	Coverage	0.930	0.910	0.860	0.780	0.780	0.960	0.940	0.950
	Coverage.bt	0.960	0.930	0.890	0.940	0.930	0.960	0.940	0.970

Standard: the linearization method based on Linstram and Bates (1990). TS: two-step method. JM: Joint model. SE.mo: model-based SE. SE.em: empirical SE from simulation. SE.bt: Bootstrap SE.

Table 5.6: Simulation results with $n_i = 30$

Model	Parameter	β_1	β_2	β_3	α_1	α_2	γ_0	γ_1	γ_2
	True value	2.5	3.0	7.5	-8	1.5	5.2	1.6	-1.2
Standard	Estimate	2.511	3.009	7.737					
	SE.mo	0.056	0.126	0.632					
	SE.em	0.063	0.096	0.705					
	rBias(%)	0.438	0.301	3.166					
	rMSE(%)	0.160	0.304	7.314					
	Coverage	0.920	0.980	0.910					
TS	Estimate	2.507	3.007	7.651	-8.199	1.524	5.204	1.596	-1.195
	SE.mo	0.045	0.060	0.338	0.354	0.062	0.041	0.047	0.046
	SE.em	0.056	0.057	0.376	0.506	0.089	0.039	0.050	0.048
	rBias(%)	0.272	0.221	2.017	2.488	1.620	0.074	0.249	0.431
	rMSE(%)	0.125	0.110	2.171	3.662	0.557	0.030	0.155	0.191
	Coverage	0.850	0.960	0.910	0.770	0.840	0.940	0.960	0.970
JM	Estimate	2.508	3.008	7.710	-8.019	1.492	5.203	1.599	-1.197
	SE.mo	0.045	0.060	0.343	0.352	0.061	0.041	0.046	0.044
	SE.em	0.056	0.058	0.396	0.568	0.100	0.039	0.049	0.047
	SE.bt	0.048	0.063	0.385	0.552	0.097	0.042	0.046	0.044
	rBias(%)	0.328	0.280	2.797	0.242	0.521	0.057	0.034	0.225
	rMSE(%)	0.125	0.114	2.660	3.990	0.665	0.030	0.150	0.184
	Coverage	0.850	0.950	0.880	0.770	0.770	0.940	0.950	0.940
	Coverage.bt	0.890	0.950	0.910	0.940	0.940	0.960	0.960	0.940

Standard: the linearization method based on Linstram and Bates (1990). TS: two-step method. JM: Joint model. SE.mo: model-based SE. SE.em: empirical SE from simulation. SE.bt: Bootstrap SE.

inferential tools for virological data from aids clinical trials. *Biometrics*, 55(2):410–418, 1999.

A Supplementary Figures for Section 4

Figure A.1: CD4 trajectories for all subjects

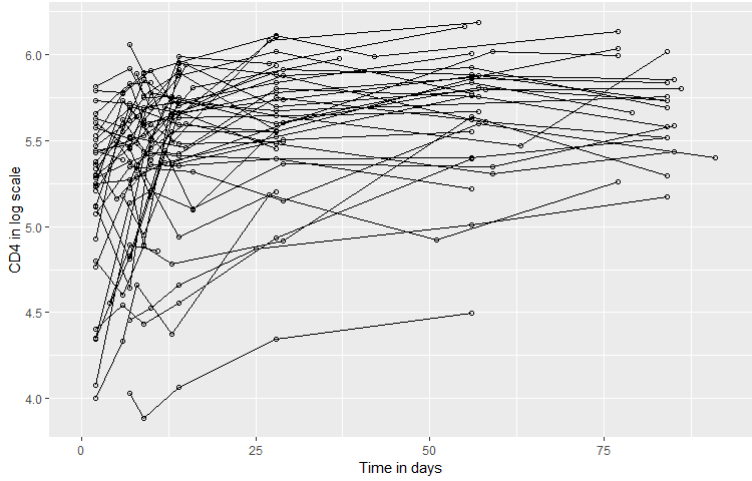
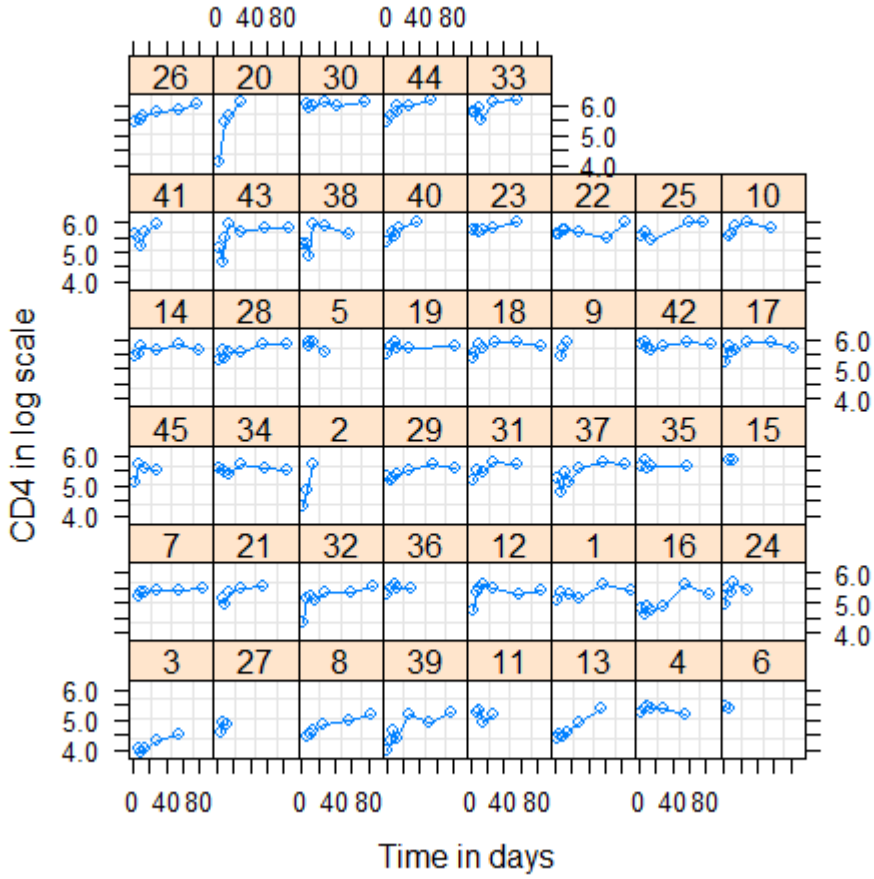



Figure A.2: CD4 trajectory for each subject